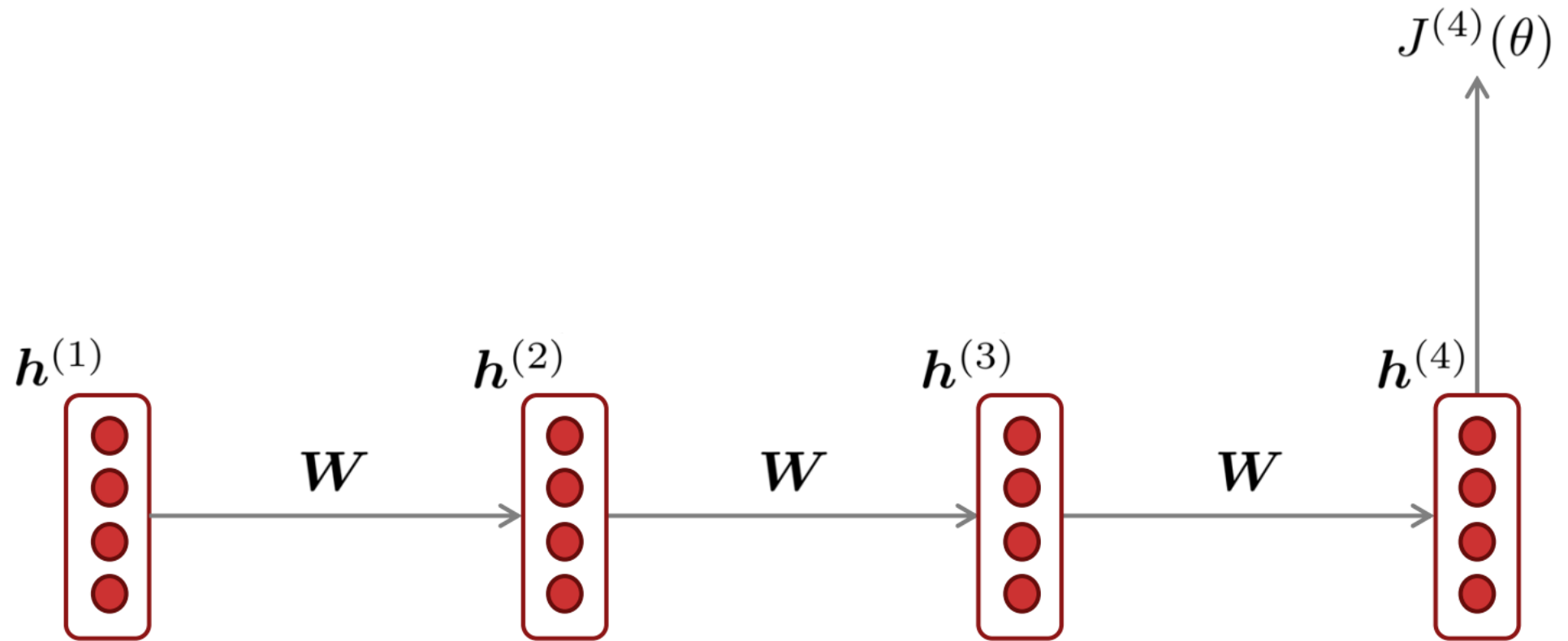
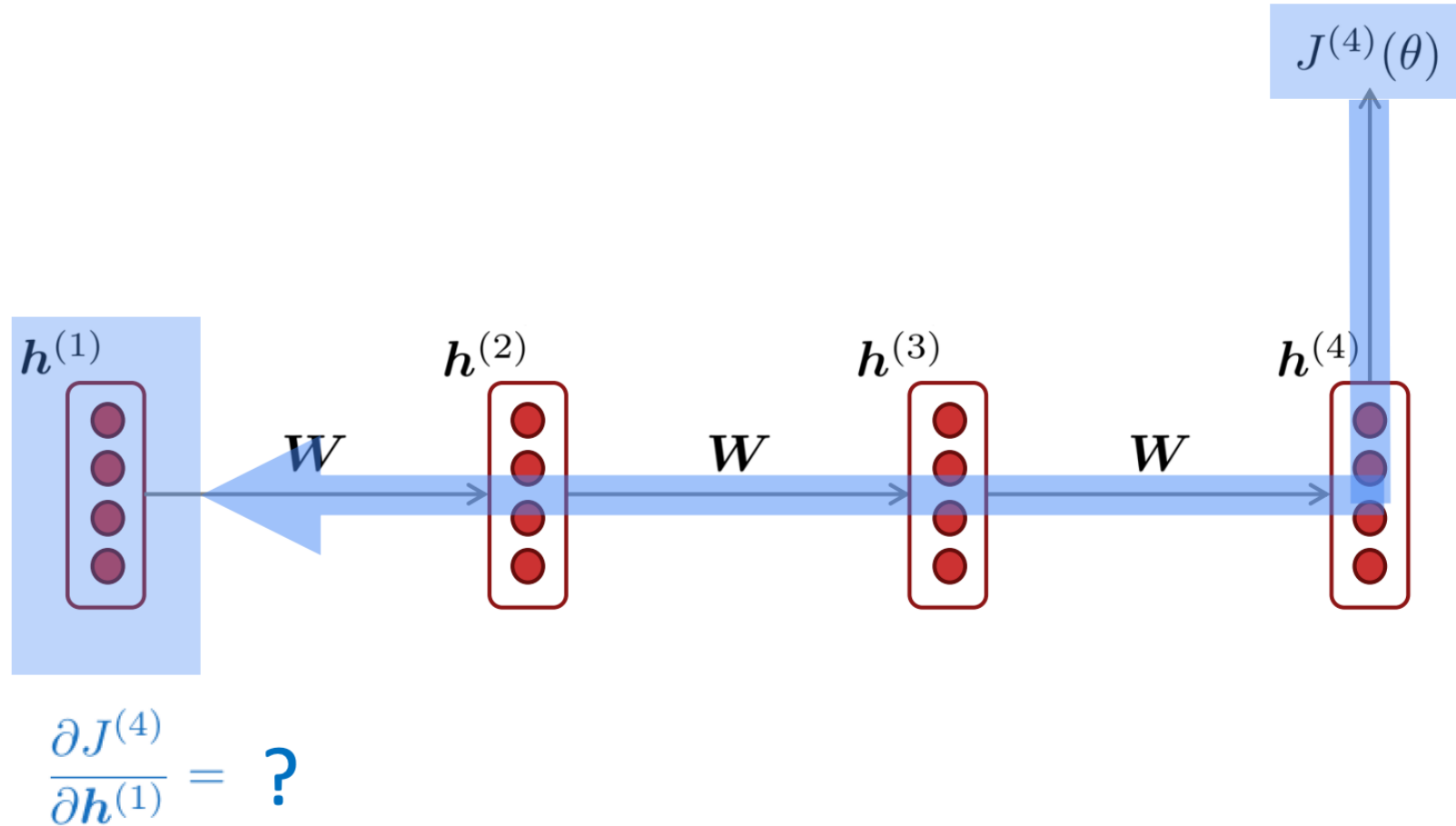


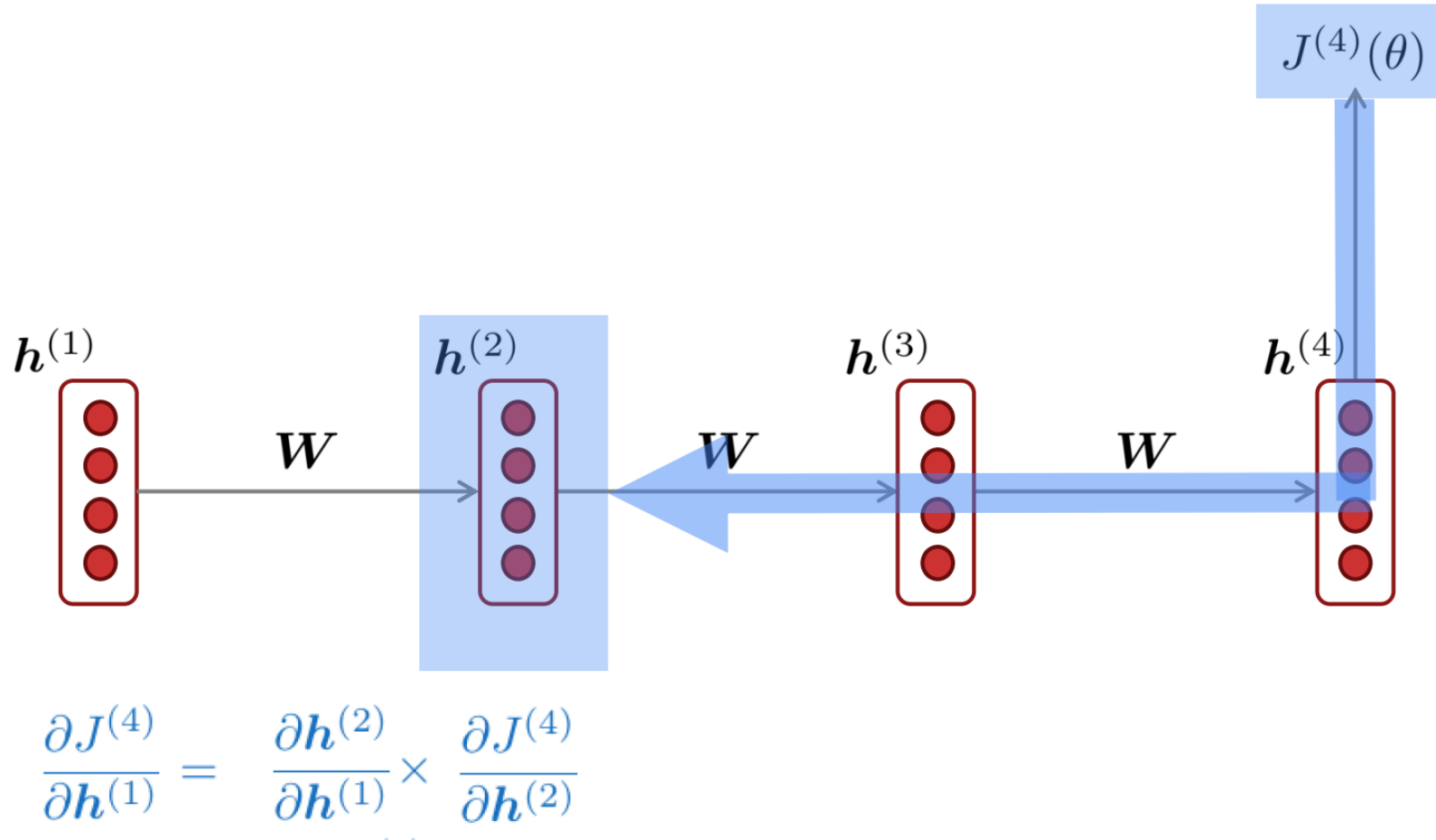
Vanishing and Exploding Gradients: Intuition

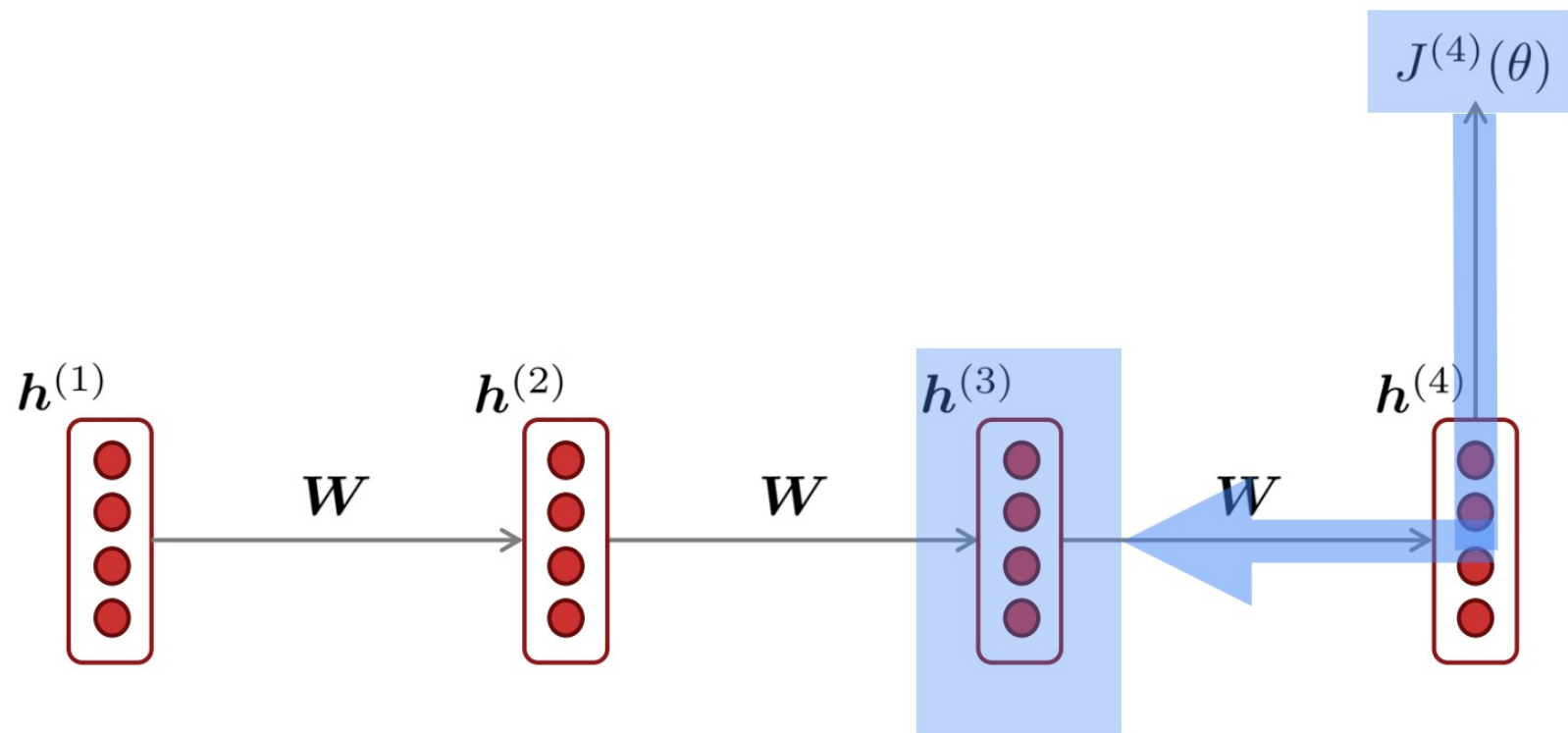


Vanishing and Exploding Gradients: Intuition



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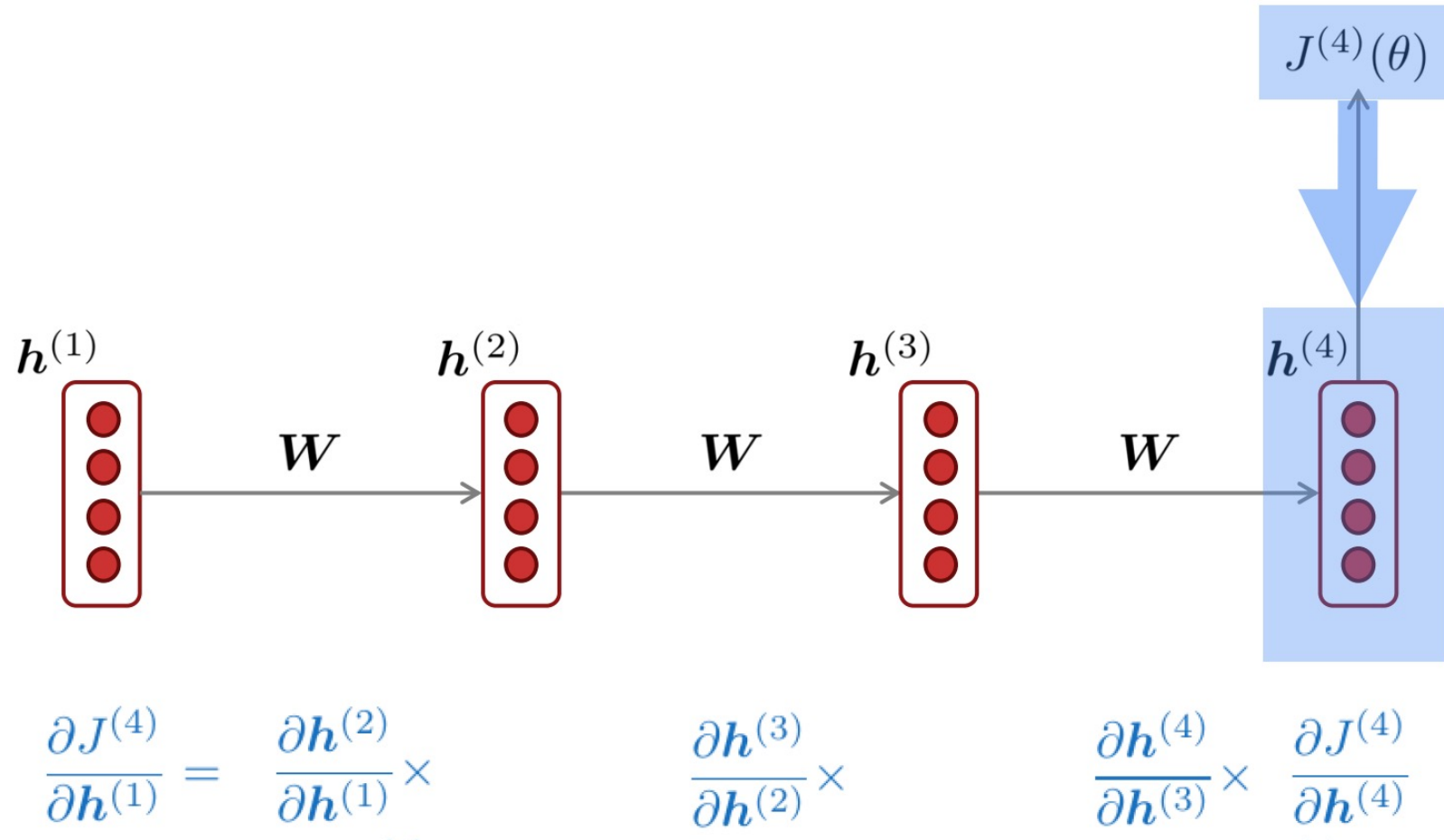


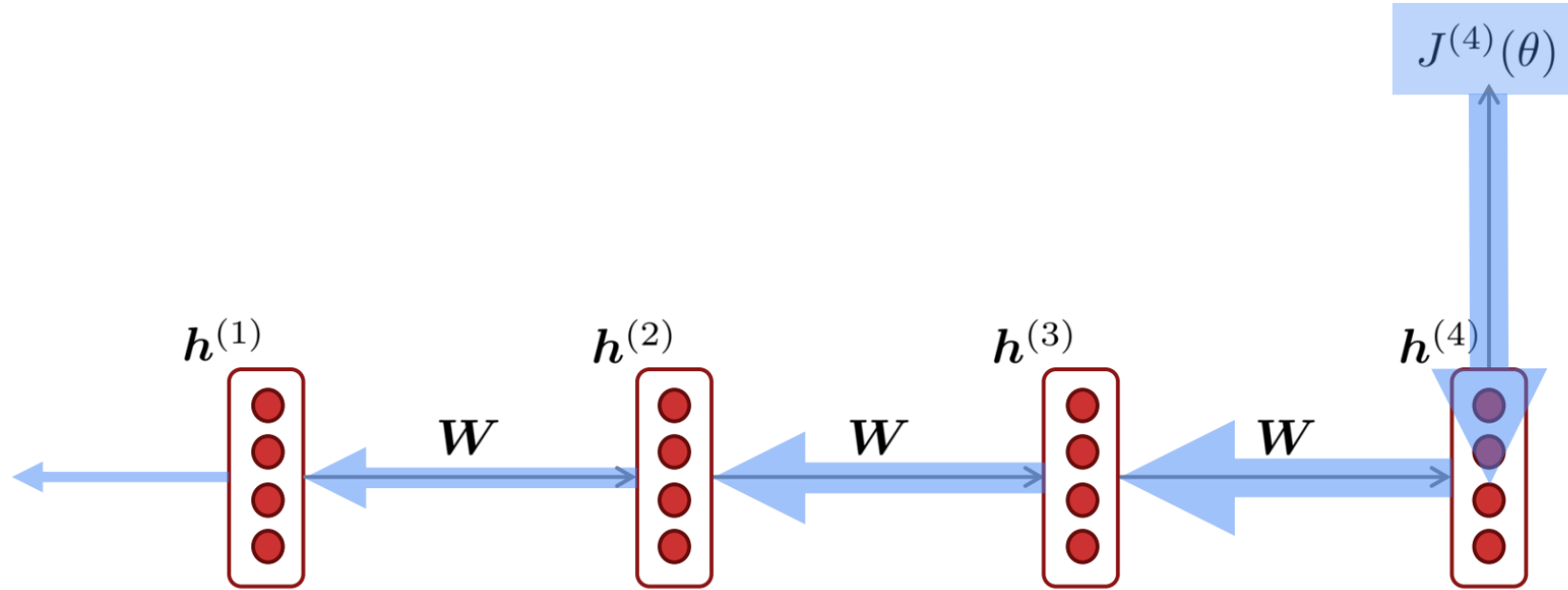


$$\frac{\partial J^{(4)}}{\partial h^{(1)}} = \frac{\partial h^{(2)}}{\partial h^{(1)}} \times$$

$$\frac{\partial h^{(3)}}{\partial h^{(2)}} \times \frac{\partial J^{(4)}}{\partial h^{(3)}}$$

Vanishing and Exploding Gradients: Intuition





$$\frac{\partial J^{(4)}}{\partial h^{(1)}} = \boxed{\frac{\partial h^{(2)}}{\partial h^{(1)}}} \times \boxed{\frac{\partial h^{(3)}}{\partial h^{(2)}}} \times \boxed{\frac{\partial h^{(4)}}{\partial h^{(3)}}} \times \frac{\partial J^{(4)}}{\partial h^{(4)}}$$

What happens if these are small?

Vanishing gradient problem:
When these are small, the gradient signal gets smaller and smaller as it backpropagates further

Vanishing Gradient Problem

- The difficulty with vanishing gradients is that it makes it very difficult for the model to learn long-distance dependencies
- The effect of changes to the state in the n -th position before, is miniscule
 - Therefore, the hidden state in the n -th position before has no causal effect on the prediction n positions ahead
- This was addressed by a different RNN architecture called LSTM (Long Short Term Memory; Hochreiter and Schmidhuber, 1997)